Immunity

Zoology

Lecturer

December 03, 2025

Outline

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Susceptibility and Resistance

- **Susceptibility** (易感性): Host cannot eliminate parasite before it becomes established.
- **Resistance** (抗性): Host’s physiological status prevents establishment and survival of parasite.
- **Immunity** (免疫): Possession of cells or tissues capable of recognizing and protecting the animal against nonself invaders.
- **Innate immunity** (先天性免疫): Defense that does not depend on prior exposure to the invader.
- **Acquired immunity** (获得性免疫): Specific to a particular nonself material, requires time for development, and occurs more quickly and vigorously on secondary response.

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Physical and Chemical Barriers

- **Physical and Chemical Barriers** (物理和化学屏障):
  - Skin (tough, cornified, or sclerotized).
  - Mucus (lubricates, dislodges particles).
  - Antimicrobial substances: Lysozyme (溶菌酶), IgA, Interferons (干扰素), Tumor necrosis factor (TNF).
  - Low pH in stomach and vagina.
  - Commensal bacteria (inhibit pathogens).
- **Complement** (补体): Series of proteins activated in sequence to lyse invading cells, tag them for phagocytosis (**opsonization**, 调理作用), and attract leukocytes.

Antimicrobial Peptides

- **Antimicrobial Peptides** (抗菌肽):
  - Defensins (防御素) in mammals.
  - Released upon recognition of microbial molecules by receptors like **Toll-like receptors (TLRs)**.

Cellular Defenses: Phagocytosis

- **Phagocytosis** (吞噬作用): Engulfing and digesting foreign particles.
- **Phagocytes** (吞噬细胞):
  - Monocytes (单核细胞) -> Macrophages (巨噬细胞).
  - Polymorphonuclear leukocytes (PMNs, 多形核白细胞) / Granulocytes (粒细胞): Neutrophils (中性粒细胞), Eosinophils (嗜酸性粒细胞), Basophils (嗜碱性粒细胞).
- **Natural killer (NK) cells** (自然杀伤细胞): Kill virus-infected and tumor cells.

Lineage of Immune Cells

- Lineage of some cells active in immune response.
- Derived from multipotential stem cells in bone marrow.
- B cells mature in bone marrow.
- T cells mature in thymus.
- Monocytes differentiate into macrophages.

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Invertebrate Immunity Overview

- Recognition of nonself (allograft/xenograft rejection).
- Phagocytosis and encapsulation.
- Antimicrobial peptides.
- Some evidence of specific memory in certain invertebrates (e.g., copepods).

Invertebrate Leukocytes

- Various invertebrate groups have specialized leukocytes.
- Functions include phagocytosis, encapsulation, and graft rejection.
- Examples: Archaeocytes in sponges, Amebocytes in molluscs.

Invertebrate Immune Response Example

- Specific examples of invertebrate immune responses include the encapsulation and phagocytosis of parasites.
- Resistant snails can attack **Schistosoma mansoni** larvae.
- Hemocytes engage in phagocytosis of the larval tegument.

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Acquired Immunity Basics

- Stimulated by **Antigen** (抗原).
- Two arms:
  - **Humoral immunity** (体液免疫): Based on **antibodies** (抗体).
  - **Cellular immunity** (细胞免疫): Associated with cell surfaces.
- **Basis of Self and Nonself Recognition**:
  - **Major Histocompatibility Complex (MHC)** (主要组织相容性复合体): Proteins on cell surfaces.
    - Class I: On virtually all cells.
    - Class II: On immune cells (lymphocytes, macrophages).

Recognition Molecules: Antibodies

- **Antibodies** (Immunoglobulins, Ig):
  - Structure: 2 heavy chains, 2 light chains, Y-shape.
  - Variable region (antigen binding), Constant region.
  - Classes: IgM, IgG, IgA, IgD, IgE.
  - Functions: Opsonization, neutralization, activation of complement, Antibody-dependent cell-mediated cytotoxicity (ADCC).

Antibody Structure

- Antibody molecule composed of light and heavy chains.
- Held together by disulfide bonds.
- Variable regions determine antigen specificity.
- Constant regions determine antibody class and function.

T-Cell Receptors and Subsets

- **T-Cell Receptors**: Transmembrane proteins on T cells, bind specific antigens.
- **Subsets of T Cells**:
  - **Helper T cells (TH)** (辅助 T 细胞): CD4 coreceptor. Secrete cytokines. TH1 (cell-mediated), TH2 (humoral).
  - **Cytotoxic T lymphocytes (CTLs)** (细胞毒性 T 细胞): CD8 coreceptor. Kill target cells.
  - **Suppressor T cells** (抑制 T 细胞).
  - **Memory T cells** (记忆 T 细胞).

Cytokines

- **Cytokines** (细胞因子): Protein hormones for communication.
- Examples: Interleukins (IL), Interferons (IFN), Tumor necrosis factor (TNF).
- Regulate immune response, inflammation, and cell proliferation.

Immune Response Pathways

- Major pathways involved in cell-mediated (TH1) and humoral (TH2) immune responses.
- Mediated by cytokines.
- TH1 activates macrophages and CTLs.
- TH2 activates B cells to produce antibodies.

Important Cytokines

- Cytokines have specific sources and functions.
- IL-1: Activates T cells, B cells, macrophages.
- IL-2: Growth factor for T and B cells.
- IL-4, IL-5, IL-6: Promote B cell growth and differentiation.
- IFN-gamma: Macrophage activating factor.

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Humoral Response Mechanism

- Antigen binds to B cell surface antibody.
- B cell internalizes and presents **epitope** (抗原决定簇) with MHC II.
- TH2 cell recognizes epitope-MHC II complex (aided by CD4).
- TH2 secretes cytokines (IL-4, IL-5, etc.) to activate B cell.
- B cell proliferates into **Plasma cells** (浆细胞) (secrete antibody) and **Memory cells** (记忆细胞).

Humoral Immune Response Diagram

- Steps in humoral immune response:
  1. Antigen binding and presentation.
  2. TH2 cell recognition and activation.
  3. B cell proliferation and differentiation.
  4. Antibody secretion by plasma cells.
  5. Opsonization of antigen.

B Cell - T Cell Interaction

- Interaction involves specific surface molecules.
- Peptide epitope presented by MHC Class II on B cell.
- T-cell receptor and CD4 on TH2 cell bind to MHC-epitope complex.
- Costimulatory molecules (CD40/CD40 receptor) are also involved.

Secondary Response

- **Secondary response** (二次免疫应答):
  - Occurs upon re-exposure to the same antigen.
  - Faster and stronger than primary response.
  - Due to presence of **Memory cells** (记忆细胞).
  - Basis for vaccination.

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Cell-Mediated Response

- Involves TH1 cells, CTLs, macrophages.
- TH1 cells recognize epitope-MHC II on APCs.
- Secrete IL-2, TNF, IFN-gamma.
- Activate CTLs (kill infected cells), NK cells, and macrophages.
- Important for intracellular parasites (viruses, bacteria) and tumor cells.
- Rejection of tissue transplants (suppressed by drugs like cyclosporine).

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Inflammation Overview

- **Inflammation** (炎症): Nonspecific response to tissue damage or infection.
- Signs: Redness, swelling, heat, pain.
- Mediators: Histamine (from mast cells), TNF, cytokines.
- Vasodilation and increased permeability allow leukocytes to enter tissue.

Delayed Type Hypersensitivity (DTH)

- **Delayed type hypersensitivity (DTH)**:
  - Cell-mediated (macrophages).
  - Takes 24 hours or more.
  - Can lead to granuloma formation around persistent antigens (e.g., parasite eggs).

Immediate Hypersensitivity

- **Immediate hypersensitivity**:
  - Antibody-mediated (IgE, mast cells).
  - Rapid release of histamine.
  - Causes wheal and flare reaction.
  - Basis of **Allergy** (过敏) and **Asthma** (哮喘).

Outline

AIDS

- Caused by **HIV** (Human Immunodeficiency Virus).
- Virus preferentially invades and destroys **Helper T cells** (TH cells) via CD4 receptor.
- Disables both humoral and cell-mediated immunity.
- Leads to susceptibility to opportunistic infections.

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ABO Blood Types

- **ABO Blood Types**: Antigens A and B on erythrocytes.
  - Type A: A antigen, anti-B antibody.
  - Type B: B antigen, anti-A antibody.
  - Type AB: A and B antigens, no antibodies (Universal recipient).
  - Type O: No antigens, anti-A and anti-B antibodies (Universal donor).

Rh Factor

- **Rh Factor**: Rh positive (has antigen) vs. Rh negative.
- **Erythroblastosis fetalis** (新生儿溶血症):
  - Rh- mother sensitized by Rh+ fetus.
  - Anti-Rh antibodies (IgG) cross placenta in subsequent pregnancy.
  - Attack fetal Rh+ red blood cells.

Major Blood Groups Table

- Summary of ABO blood groups.
- Shows genotypes, antigens, antibodies, and compatibility.
- Frequencies vary among populations.